

Team Hawkeye at Oleekbochon: A Hybrid Program-Aided Reasoning and RAG-Augmented Fine-Tuning Framework for Bengali LLM Hallucination Detection*

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Abstract

Large Language Models frequently generate fluent yet factually incorrect statements—a problem especially acute in low-resource languages such as Bengali. We present Team Hawkeye’s solution for the *Oleekbochon Bengali LLM Hallucination Detection Challenge*. Our framework integrates: (1) a Self-Consistency Thought-augmented Reasoning (SC-TIR) code execution engine with Bengali numeric parsing for mathematical queries; (2) a dense-sparse RAG pipeline using BGE-M3 bi-encoders and cross-encoder rerankers over a 250k+ chunk Bengali Wikipedia corpus; (3) probabilistic synthetic data generation (100k samples) across four hallucination categories; and (4) a fine-tuned Qwen3.5-4B log-prob classifier via vLLM. Our system achieved **0.83834 Macro F1** (13th place), improving **+16 ranks** over Phase 1.

Keywords: Hallucination Detection, Bengali NLP, RAG, Program-Aided Language Models, SFT, Oleekbochon.

1 Introduction

Large Language Models have achieved remarkable NLP advancements, yet their tendency to generate plausible but factually inaccurate assertions—*hallucination*—remains a fundamental deployment obstacle in high-stakes domains such as education, law, and healthcare (Ji et al., 2023). While extensive research has focused on hallucination detection in English, low-resource and morphologically complex languages face severe vulnerabilities.

Bengali, spoken by over 270 million people worldwide, presents unique computational challenges for LLM factual verification: **(1) Complex Numeric Representations:** Numbers can

appear as ASCII digits (71), Bengali script digits, or spelled-out word forms (*Ekattor*), and standard tokenizers often segment Bengali subwords inconsistently, leading to arithmetic matching failures; **(2) Culturally Situated Knowledge Deficit:** Generic LLMs pre-trained on predominantly Western text corpora exhibit frequent entity swaps when prompted about Bengali literature, history, and regional geography; and **(3) Idiomatic Ambiguity:** Bengali proverbs and idioms (*bagdhara*) often have non-literal meanings (e.g., *Shibratrir Solte* referring to an only surviving offspring rather than a literal lamp), causing LLMs to generate incorrect interpretations.

The **Oleekbochon Challenge** (Islamic University of Technology Computer Society (IUTCS), 2026), hosted by IUTCS, tasks participants with binary faithfulness prediction for Bengali question-response pairs under Macro F1 evaluation. Our contributions are:

1) Program-Aided Math Reasoning: A deterministic math query router, Python code generation solver, sandboxed executor, and Bengali word-to-number converter for resolving mathematical claims.

2) Dense-Sparse RAG: A high-throughput retrieval engine over 250k+ Bengali Wikipedia chunks using BGE-M3 bi-encoders and BGE-Reranker-v2 cross-encoders.

3) Probabilistic Synthetic SFT: 100,000 training pairs across four hallucination categories with K -chunk distractor context mixing.

4) Fine-Tuned Verification: Qwen3.5-4B with LoRA, deployed via vLLM with Softmax-normalized logprob classification.

Our framework propelled Team Hawkeye from 29th to **13th place** (+16 ranks) with **0.83834 Macro F1**.

*Technical report for the Oleekbochon Bengali LLM Hallucination Detection Challenge organized by IUTCS.

2 Related Work

2.1 Hallucination Detection

Detection methods fall into self-consistency metrics, NLI entailment scoring, and reference-based retrieval (Manakul et al., 2023; Es et al., 2023). SelfCheckGPT evaluates factual consistency by sampling multiple stochastic outputs and computing token-level entropy without external references. NLI-based systems frame detection as premise-hypothesis entailment, though standard cross-encoders struggle on non-English text.

2.2 Program-Aided Language Models

PAL (Gao et al., 2023) offloads intermediate reasoning to code interpreters, decoupling logic synthesis from arithmetic evaluation. We adapt PAL for Bengali by introducing custom keyword heuristics and numeric script converters.

2.3 Dense Retrieval for Low-Resource Languages

RAG enhances factual grounding by injecting relevant passage context (Lewis et al., 2020). BGE-M3 (Chen et al., 2024) supports multi-linguality across dense, sparse, and multi-vector representations. Combining bi-encoder retrieval with cross-encoder reranking significantly improves precision in low-resource search.

3 System Architecture

Our system (Fig. 1) operates as a multi-stage pipeline: queries are first routed through the SC-TIR Math module; remaining queries are enriched via the RAG engine, processed by the SFT classifier, and ensembled into the final submission.

3.1 Stage 1: SC-TIR Math Verification Engine

Bengali math word problems (e.g., age ratios, interest rates) are highly susceptible to tokenization errors. We design a dedicated Program-Aided Reasoning pipeline.

3.1.1 Math Query Routing

A scoring function $\mathcal{S}_{\text{math}}(q)$ is computed using Bengali digit occurrences, arithmetic operators, ratio patterns, and domain keywords ($\mathcal{K}_{\text{math}}$: *onupat*, *gawd*, *shotkora*, *labh*, *khoti*, *taka*):

$$\begin{aligned} \mathcal{S}_{\text{math}}(q) = & w_d \cdot \mathbb{I}(q \text{ has digits}) \\ & + \sum_{k \in \mathcal{K}_{\text{math}}} w_k \cdot \mathbb{I}(k \in q) \\ & - w_h \cdot \mathbb{I}(q \text{ has date terms}) \end{aligned} \quad (1)$$

Algorithm 1 SC-TIR Math Verification Pipeline

Require: Query q , Response r , Model $\mathcal{M}_{\text{code}}$, Tolerance ϵ

Ensure: Binary Label $y \in \{0, 1\}$ or Fallback

```

1:  $s \leftarrow \mathcal{S}_{\text{math}}(q)$ 
2: if  $s < 5$  then
3:   return FALLBACK_TO_RAG
4: end if
5:  $\text{Code} \leftarrow \mathcal{M}_{\text{code}}.\text{Generate}(\text{BuildSCPrompt}(q))$ 

6:  $\hat{y} \leftarrow \text{SandboxedExecute}(\text{Code}, 5s)$ 
7: if  $\hat{y}$  is NULL then
8:   return FALLBACK_TO_LLM_JUDGE
9: end if
10:  $\mathcal{C} \leftarrow \text{ExtractNumericCandidates}(r)$ 
11: if  $\text{ValuesMatch}(\hat{y}, \mathcal{C}, \epsilon)$  then
12:   return  $y = 1$ 
13: else
14:   return  $y = 0$ 
15: end if
```

Queries with $\mathcal{S}_{\text{math}}(q) \geq 5$ are routed to the code executor.

3.1.2 Code Generation and Execution

Routed queries are sent to Qwen3-8B via vLLM to generate Python code saving the result to `final_answer`. Code executes inside an isolated multiprocessing worker with a 5-second timeout.

3.1.3 Bengali Numeric Matching

Answers are clustered via tolerance matching:

$$|v_1 - v_2| \leq \max(\epsilon_{\text{abs}}, \epsilon_{\text{rel}} \cdot \max(|v_1|, |v_2|)) \quad (2)$$

where $\epsilon_{\text{rel}} = 10^{-3}$, $\epsilon_{\text{abs}} = 10^{-2}$. A deterministic Bengali number word parser converts units (0–99) and scale multipliers (*shot*=100, *hajar*=1000, *lakh*= 10^5 , *koti*= 10^7) to numeric values. If the computed answer $\hat{y}$ matches any candidate in the response, the label is Faithful; otherwise Hallucinated. Algorithm 1 formalizes the pipeline.

3.2 Stage 2: Dense-Sparse RAG Engine

Non-mathematical queries require external evidence for factual verification.

3.2.1 Knowledge Base Construction

We construct a Bengali knowledge base aggregating 12 multi-domain datasets: TituLM Bangla MMLU (Nahin et al., 2025), Bengali Wikipedia,

Table 1: Empirical Pipeline Evaluation across LLM Backbones and Verification Paradigms

Pipeline Architecture	Backbone LLM	Training & Feature Configuration	Public F1	Private F1
Direct SFT Classifier (Best)	Qwen3.5-4B	5k Augmented + RAG + SC-TIR Math	0.901	0.903
Direct SFT Classifier (9B Scale)	Qwen3.5-9B	10k Augmented + RAG + SC-TIR Math	0.899	0.894
Untrained Few-Shot Baseline	Gemma-4-26B	Few-Shot + RAG + SC-TIR Math	0.894	0.883
SFT Hunter + Fine-Tuned NLI Judge	Qwen3.5-4B	5k Augmented + Fine-Tuned NLI Judge	0.868	0.876
SFT Hunter + Pretrained NLI	Qwen3.5-4B	5k Augmented + Pretrained NLI Classifier	0.869	0.859
Few-Shot Baseline (No RAG/Math)	Gemma-31B	Few-Shot (Without RAG & Math Solver)	0.834	0.843
SFT on Raw Synthetic (100k)	Qwen3.5-9B	100k Un-augmented SFT (qwen_train) + RAG	0.800	0.795
One-Shot Baseline (No RAG/Math)	Gemma-31B	1-Shot (Without RAG & Math Solver)	0.694	0.683

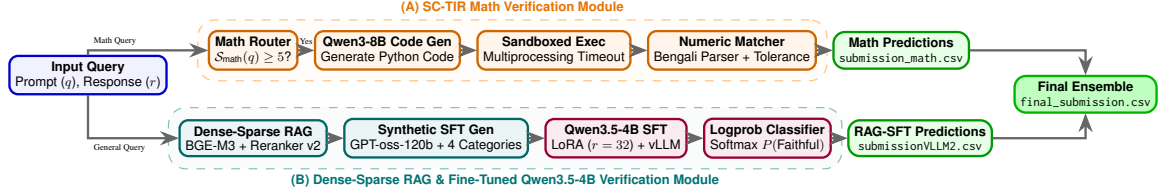


Figure 1: Overview of Team Hawkeye’s Multi-Stage Hallucination Verification Framework.

Bangla Bagdhara (Adib et al., 2026b), BenHalluEval (Adib et al., 2026a), BLAD (Sakhawat, 2025), NCTB textbooks (Class 8–10), Bangladesh Open University books, BLUCK benchmark (Kabir et al., 2025), Government Bangladesh Lexicon (Government of Bangladesh, 2024), BanglaVerb (Ayman et al., 2025b), BanglaBlend (Ayman et al., 2025a), and BanglaTense (Bijoy et al., 2024). Raw texts are NFKC-normalized and chunked into passages of 500 characters with 100-character overlap using sentence-aware boundary detection, yielding 250,000+ indexed chunks.

3.2.2 Dense Indexing and Reranking

Passages are embedded into 1024-dim vectors using BAAI/bge-m3 (FP16, batch size 2048) and L_2 -normalized:

$$\mathbf{v}_p = \frac{\mathbf{e}(p)}{\max(\|\mathbf{e}(p)\|_2, 10^{-12})} \quad (3)$$

Embeddings are indexed into FAISS Flat Inner Product (IndexFlatIP) in blocks of 100k vectors, ensuring 100% recall. For each query q , the top $M=40$ candidate passages are retrieved:

$$\mathcal{P}_{\text{topM}} = \arg \max_{p \in \text{KB}}^{(M)} \left(\mathbf{v}_q^\top \mathbf{v}_p \right) \quad (4)$$

The query-passage pairs (q, p_i) are then reranked using BAAI/bge-reranker-v2-m3; the top $K=5$ reranked passages are extracted, formatted with reference indices, and concatenated into the retrieved context.

3.3 Stage 3: Synthetic Data and SFT Augmentation

To fine-tune a robust classifier without relying solely on limited ground-truth data, we create a large synthetic dataset using openai/gpt-oss-120b via the Tinker API.

3.3.1 Hallucination Category Generation

Given factual Bengali context snippets, the teacher model generates four error categories:

1. **Entity Swap:** Swapping names, locations, or organizations with related terms.
2. **Numeric Alteration:** Modifying dates, quantities, or counts.
3. **Contradiction:** Inverting the logical truth of a factual claim.
4. **Baseless Addition:** Introducing realistic yet unmentioned details.

3.3.2 Probabilistic Context-Mixing

To prevent positional bias, we apply four mixing operations: baseline clean context (30%), single distractor swap with fallback token [Information Not Found] (15%), target mixed with $K-1$ distractors in shuffled order (40%), and distractors-only without target (15%). From 100,000 augmented samples, 5,000 balanced samples are selected for fine-tuning.

3.4 Stage 4: Verification Model and Inference

3.4.1 SFT Information Hunter

We train Qwen3.5-4B via LoRA ($r=32$, $\eta=2 \times 10^{-4}$) as an *Information Hunter*: given

question q and RAG context, the model extracts supporting evidence verbatim or returns the fallback token [Information Not Found] (*tottho paowa jayni*).

3.4.2 NLI Verification Backends

Once evidence e is extracted, we evaluate two NLI paradigms: (1) a fine-tuned cross-encoder judge accepting (Premise= e , Hypothesis= r) for binary entailment, and (2) an off-the-shelf multilingual NLI classifier as baseline.

3.4.3 Logprob Classification and Inference

For direct verification, candidate logprobs $\mathcal{L}_{\text{true}}$ and $\mathcal{L}_{\text{false}}$ are Softmax-normalized:

$$P(\text{Faithful}) = \frac{\exp(\mathcal{L}_{\text{true}})}{\exp(\mathcal{L}_{\text{true}}) + \exp(\mathcal{L}_{\text{false}})} \quad (5)$$

Prompts are dynamically truncated via `fit_retrieved_context_to_budget` for Tesla T4 VRAM limits. Final predictions combine SC-TIR math and SFT verification outputs.

4 Experimental Setup and Results

4.1 Dataset and Evaluation Metric

The competition test dataset consists of 2,598 test rows containing Bengali questions, candidate responses, and optional context snippets. Performance is evaluated strictly using Macro F1-score:

$$\text{Macro F1} = \frac{\text{F1}_{\text{Faithful}} + \text{F1}_{\text{Hallucinated}}}{2} \quad (6)$$

4.2 Official Competition Standings

Table 2 presents the official Phase 2 standings. Team Hawkeye achieved a final Macro F1-score of **0.83834**, securing **13th place** out of 32 qualified teams.

4.3 Phase 1 vs. Phase 2 Rank Progression

A key highlight is our system’s robustness on held-out evaluation. In the Phase 1 Private Leaderboard, our baseline model ranked **29th place**. Following the introduction of our SC-TIR math solver, BGE-M3 RAG reranking, and probabilistic synthetic SFT augmentation, we climbed **+16 positions** to **13th place** in the final standings.

4.4 Empirical Pipeline Comparison and Comparative Analysis

To systematically evaluate our architectural design choices, we conduct extensive empirical experiments across multiple

Table 2: Official Phase 2 Final Standings (Oleekbochon Challenge)

Rank	Team Name	Macro F1-Score
1	BUET_NeuralX	0.95424
2	Team_ReActive	0.92392
3	DU_RDanto	0.92120
...	...	...
13	Hawkeye (Ours)	0.83834

LLM backbones (Qwen3.5-4B, Qwen3.5-9B, Gemma-4-26B, Gemma-4-31B) and verification paradigms (Direct SFT Logprob Classification vs. Fine-Tuned NLI Judge vs. Pretrained NLI Classifier vs. Un-augmented SFT). Table 1 summarizes the Public and Private test set performance across all evaluated system configurations.

1) Superiority of Direct Logprob Classification: Our fine-tuned Qwen3.5-4B model trained on 5,000 synthetic samples with probabilistic distractor mixing achieved the highest overall score of **0.901 Public F1** and **0.903 Private F1** when evaluated directly via logprob Softmax normalization.

2) Impact of Probabilistic Distractor Mixing (+0.108 F1 Jump): A crucial finding lies in comparing direct fine-tuning on raw synthetic data versus probabilistic distractor mixing. Training Qwen3.5-9B directly on 100,000 raw un-augmented synthetic pairs generated by `generate_synthetic_data.ipynb` via `qwen_train.ipynb` achieved only **0.800 Public / 0.795 Private F1**. Without distractor mixing, the model overfit to single-passage context layouts and learned keyword shortcuts. Introducing probabilistic K -chunk distractor context mixing (AUGEMENT synthetic_sft_dataset.ipynb) propelled accuracy by **+0.108 F1 points** to **0.903 Private F1**, proving that context mixing is essential for multi-reference factual verification.

3) The “Information Hunter” Task Misalignment Paradox: A particularly notable phenomenon emerged regarding the *Information Hunter* paradigm. The Qwen3.5-4B model was specifically fine-tuned to extract supporting evidence verbatim or return [Information Not Found] (*tottho paowa jayni*). However, when its extracted evidence was passed downstream to a dedicated fine-tuned NLI judge (`train_nli_judge.ipynb`), the pipeline achieved **0.876 Private F1**—performing *worse* than using the Information Hunter LLM directly as a direct logprob classifier (**0.903 Private F1**). The model developed superior internal representations for di-

rect factual classification while learning evidence extraction than when used for the explicit downstream extraction task.

4) Ablation of RAG Retrieval and SC-TIR Math Solver: Omitting both the Dense-Sparse RAG engine and SC-TIR Math Solver caused severe performance degradation. An un-augmented few-shot model scored **0.843 Private F1**, while a one-shot model dropped to **0.683 Private F1**, demonstrating that external factual grounding and deterministic code execution are indispensable for Bengali LLM hallucination detection.

5 Conclusion and Future Work

In this paper, we presented Team Hawkeye’s 13th-place solution for the *Oleekbochon Bengali LLM Hallucination Detection Challenge*. By integrating Program-Aided Language reasoning for math claims, dense-sparse BGE-M3 retrieval over a 250k+ passage Wikipedia index, probabilistic synthetic SFT data generation across four hallucination categories, and fine-tuned Qwen3.5-4B logprob classification, our framework demonstrated high accuracy and robustness with a +16 leaderboard rank progression.

A key empirical insight was that direct logprob classification outperformed the decoupled Information Hunter + NLI Judge pipeline, suggesting that intermediate evidence extraction may introduce information bottlenecks when the LLM has already internalized strong factual representations during SFT.

Future work should explore: (1) scaling to larger backbone models such as Gemma-4 31B within our pipeline; (2) incorporating chain-of-thought self-verification for improved coverage on edge cases; and (3) extending the framework to other low-resource South Asian languages facing similar hallucination challenges.

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